

CB 3.44 For any random variable X for which EX^2 and $E|X|$ exist, show that $P(|X| \geq b)$ does not exceed either EX^2/b^2 or $E|X|/b$, where b is a positive constant. If $f(x) = e^{-x}$ for $x > 0$, show that one bound is better when $b = 3$ and the other when $b = \sqrt{2}$. (Notice Markov's Inequality in Miscellanea 3.8.2.)

The function $g(x) = |x|$ is a nonnegative function. So by Chebychev's Inequality,

$$P(|X| \geq b) \leq E|X|/b$$

Also, $P(|X| \geq b) = P(X^2 \geq b^2)$. Since $g(x) = x^2$ is also nonnegative, again by Chebychev's Inequality we have

$$P(|X| \geq b) = P(X^2 \geq b^2) \leq EX^2/b^2.$$

For $X \sim \text{exponential}(1)$, $E|X| = EX = 1$ and $EX^2 = \text{Var}X + (EX)^2 = 2$. For $b = 3$,

$$E|X|/b = 1/3 > 2/9 = EX^2/b^2.$$

Thus EX^2/b^2 is a better bound. But for $b = \sqrt{2}$,

$$E|X|/b = 1/\sqrt{2} < 1 = EX^2/b^2.$$

Thus $E|X|/b$ is a better bound.

CB 3.45 Let X be a random variable with moment-generating function $M_X(t)$, $-h < t < h$.

- (a) Prove that $P(X \geq a) \leq e^{-at}M_X(t)$, $0 < t < h$. (A proof similar to that used for Chebychev's Inequality will work.)

$$\begin{aligned} M_X(t) &= \int_{-\infty}^{\infty} e^{tx} f_X(x) dx \geq \int_a^{\infty} e^{tx} f_X(x) dx \\ &\geq e^{ta} \int_a^{\infty} f_X(x) dx = e^{ta} P(X \geq a), \end{aligned}$$

where we use the fact that e^{tx} is increasing in x for $t > 0$.

CB 3.46 Calculate $P(|X - \mu| \geq k\sigma)$ for $X \sim \text{uniform}(0,1)$ and $X \sim \text{exponential}(\lambda)$, and compare your answers to the bounds from Chebychev's Inequality for various values of k .

For $X \sim \text{uniform}(0, 1)$, $\mu = \frac{1}{2}$ and $\sigma^2 = \frac{1}{12}$, thus

$$P(|X - \mu| > k\sigma) = 1 - P\left(\frac{1}{2} - \frac{k}{\sqrt{12}} \leq X \leq \frac{1}{2} + \frac{k}{\sqrt{12}}\right) = \begin{cases} 1 - \frac{2k}{\sqrt{12}} & k < \sqrt{3}, \\ 0 & k \geq \sqrt{3}. \end{cases}$$

For $X \sim \text{exponential}(\lambda)$, $\mu = \lambda$ and $\sigma^2 = \lambda^2$, thus

$$P(|X - \mu| > k\sigma) = 1 - P(\lambda - k\lambda \leq X \leq \lambda + k\lambda) = \begin{cases} 1 + e^{-(k+1)} - e^{k-1} & k \leq 1 \\ e^{-(k+1)} & k > 1. \end{cases}$$

From Example 3.6.2, Chebychev's Inequality gives the bound $P(|X - \mu| > k\sigma) \leq 1/k^2$.

Comparison of probabilities			
k	u(0,1) exact	exp(λ) exact	Chebychev
.1	.942	.926	100
.5	.711	.617	4
1	.423	.135	1
1.5	.134	.0821	.44
$\sqrt{3}$	0	0.0651	.33
2	0	0.0498	.25
4	0	0.00674	.0625
10	0	0.0000167	.01

CB 5.21 What is the probability that the larger of two continuous iid random variables will exceed the population median? Generalize this result to samples of size n .

Let m denote the median. Then, for general n we have

$$\begin{aligned} P(\max(X_1, \dots, X_n) > m) &= 1 - P(X_i \leq m \text{ for } i = 1, 2, \dots, n) \\ &= 1 - (P(X_1 \leq m))^n \\ &= 1 - \left(\frac{1}{2}\right)^n \end{aligned}$$

CB 5.24 Let $X_1, \dots, X_n$ be a random sample from a population with pdf

$$f_X(x) = \frac{1}{\theta} \text{ for } 0 < x < \theta \text{ and } 0 \text{ otherwise.}$$

Let $X_{(1)} < \dots < X_{(n)}$ be the order statistics. Show that $X_{(1)}/X_{(n)}$ and $X_{(n)}$ are independent random variables.

For $f_x(x) = 1/\theta$, $F_x(x) = x/\theta$. Let $Y = X_{(n)}$ and $Z = X_{(1)}$. Then we have

$$\begin{aligned} f_{Z,Y}(z,y) &= \frac{n!}{0!(n-2)!0!} \frac{1}{\theta} \frac{1}{\theta} \left(\frac{z}{\theta}\right)^0 \left(\frac{y-z}{\theta}\right)^{n-2} \left(1 - \frac{y}{\theta}\right)^0 \\ &= \frac{n(n-1)}{\theta^n} (y-z)^{n-2}, \quad 0 < z < y < \theta \end{aligned}$$

Now we let $W = Z/Y$ and $Q = Y$. Then $Y = Q$, $Z = WQ$, and $|J| = q$ so

$$\begin{aligned} f_{W,Q}(w,q) &= \frac{n(n-1)}{\theta^n} (q - wq)^{n-2} q \\ &= \frac{n(n-1)}{\theta^n} (1-w)^{n-2} q^{n-1}, \quad 0 < w < 1; 0 < q < \theta. \end{aligned}$$

Since the pdf factors into function of w and q , W and Q are independent.

(6) $F_x(x) = \int_0^x 2t dt = t^2 \Big|_0^x = x^2$
 $f_{X_{(n)}}(x) = n[x^2]^{n-1} 2x = 2nx^{2n-2} = 2nx^{2n-1} \quad 0 < x < 1$
 $E(X_{(n)}) = \int_0^1 x \cdot 2nx^{2n-1} dx = 2n \int_0^1 x^{2n} dx$
 $= 2n \cdot \frac{1}{2n+1} x^{2n+1} \Big|_0^1 = \boxed{\frac{2n}{2n+1}}$

(7) $f_{X_{(n)}, X_{(1)}}(x_n, x_1) = n(n-1)(x_n - x_1)^{n-2} \quad 0 < x_1 < x_n < 1$

$R = X_{(n)} - X_{(1)} \quad W = X_{(1)}$

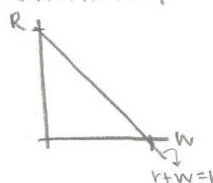
$X_{(1)} = W \quad X_{(n)} = R + W \quad 0 < W < R + W < 1$

$J=1$

$f_{R,W}(r,w) = n(n-1)(r+w-w)^{n-2}$

$f_R(r) = \int_0^{1-r} n(n-1)r^{n-2} dw$

$= n(n-1)r^{n-2}(1-r) \quad 0 < r < 1$



$$(8) f_X(x) = p^x (1-p)$$

$$F_X(x) = \sum_{k=0}^x p^k (1-p) = (1-p) \left(\frac{1-p^{x+1}}{1-p} \right) = 1 - p^{x+1}$$

$$F_{X(n)}(x) = 1 - [1 - F_X(x)]^n = 1 - [1 - (1 - p^{x+1})]^n = 1 - p^{x+1n}$$

$$\begin{aligned} f_{X(n)}(x) &= F_{X(n)}(x) - F_{X(n)}(x-1) \\ &= 1 - p^{x+1n} - (1 - p^{xn}) = 1 - p^{x+1n} + p^{xn} = p^{xn} - p^{x+1n} \\ &= p^{xn} (1 - p^n) \quad x=0, 1, 2, \dots \end{aligned}$$

$$(9) F_X(x) = x^2 \quad 0 < x < 1$$

$$\begin{aligned} (a) F_{X(3), X(4)}(x_3, x_4) &= \frac{4!}{2!} (2x_3)(2x_4) (x_3^2)^2 \quad 0 < x_3 < x_4 < 1 \\ &= 48 x_3^5 x_4 \quad 0 < x_3 < x_4 < 1 \end{aligned}$$

$$\begin{aligned} (b) f_{X(4)}(x_4) &= (2)(4)(x_4)^{(2)(4)-1} \quad (\text{from problem 3}) \\ &= 8x_4^7 \quad 0 < x_4 < 1 \end{aligned}$$

$$f_{X(3)|X(4)} = \frac{48 x_3^5 x_4}{8 x_4^7} = \frac{6 x_3^5}{x_4^6} \quad 0 < x_3 < x_4$$

$$\begin{aligned} (c) E(X(3) | X(4) = x_4) &= \int_0^{x_4} \frac{x_3 \cdot 6 x_3^5}{x_4^6} dx_3 = \frac{6}{x_4^6} \int_0^{x_4} x_3^6 dx_3 \\ &= \frac{6}{x_4^6} \cdot \frac{1}{7} (x_3^7) \Big|_0^{x_4} = \frac{6}{7} \frac{x_4^7}{x_4^6} = \left(\frac{6}{7}\right) x_4 \end{aligned}$$

10. (a) $f_X(x) = e^{-x}$ $F_X(x) = 1 - e^{-x}$ 

we want the pdf of the minimum

$$f_{X_{(1)}}(x) = n[1 - F(x)]^{n-1} f_X(x) = n[1 - 1 + e^{-x}]^{n-1} e^{-x} = ne^{-x(n-1)-x}$$

$$= he^{-nx}, \quad x_{(1)} > 0$$

(b.)  we want pdf of maximum

$$f_{X_{(n)}}(x) = n[F_X(x)]^{n-1} f_X(x) = n[1 - e^{-x}]^{n-1} e^{-x}$$

$$= he^{-x}(1 - e^{-x})^{n-1}, \quad x_{(n)} > 0$$

(c.) Use thm 5.4.4 to find pdf of the 3rd order statistic
 $n=5$ $j=3$

$$f_{X_{(3)}}(x) = \frac{5!}{(3-1)!(5-3)!} e^{-x} (1 - e^{-x})^{3-1} (e^{-x})^{5-3}$$

$$= 30e^{-3x} (1 - e^{-x})^2, \quad x_{(3)} > 0$$

(d.) $f_{X_i}(x) = \frac{1}{\theta_i} e^{-x/\theta_i}$ $F_{X_i}(x) = 1 - e^{-x/\theta_i}$

$$F_{X_{(1)}}(x) = P(X_{(1)} \leq x) = 1 - P(X_{(1)} > x) = 1 - P(\text{all } x > x)$$

$$= 1 - [P(X_1 > x) P(X_2 > x) \cdots P(X_n > x)]$$

$$= 1 - [1 - F_{X_1}(x)][1 - F_{X_2}(x)] \cdots [1 - F_{X_n}(x)]$$

$$= 1 - [(e^{-x/\theta_1})(e^{-x/\theta_2})(e^{-x/\theta_3}) \cdots (e^{-x/\theta_n})]$$

$$= 1 - e^{-x/\sum \theta_i}$$

$$f_{X_{(1)}}(x) = \frac{1}{\sum \theta_i} e^{-x/\sum \theta_i}, \quad x_{(1)} > 0$$

(11) For $X \sim \text{unif}(0,1)$, the population median $= \frac{1}{2}$

In class, we showed that $X_{(j)} \sim \text{Beta}(j, n-j+1)$

$$\text{so } E(X_{(j)}) = \frac{j}{j+n-j+1} = \frac{j}{n+1} \quad n - \frac{n+1}{2} + 1$$

n odd

The median is the $(\frac{n+1}{2})^{\text{th}}$ order stat so $j = \frac{n+1}{2}$

$$E\left(X_{(\frac{n+1}{2})}\right) = \frac{\frac{n+1}{2}}{n+1} = \frac{1}{2}$$

n even

The median is $\frac{1}{2} (X_{(\frac{n}{2})} + X_{(\frac{n}{2}+1)})$

$$E\left[\frac{1}{2} (X_{(\frac{n}{2})} + X_{(\frac{n}{2}+1)})\right] = \frac{1}{2} E(X_{(\frac{n}{2})}) + \frac{1}{2} E(X_{(\frac{n}{2}+1)})$$

$$= \left(\frac{1}{2}\right) \left(\frac{\frac{n}{2}}{n+1}\right) + \left(\frac{1}{2}\right) \left(\frac{\frac{n}{2}+1}{n+1}\right)$$

$$= \frac{1}{2} \left[\frac{\frac{n}{2} + \frac{n}{2} + 1}{n+1} \right] = \frac{1}{2} \left[\frac{n+1}{n+1} \right] = \frac{1}{2}$$

12. They just need to try to show this to get full points.

13. They just need to try to show this to get full points.

14.

$$(9.) \quad f_x(x) = e^{-(x-\mu)} \quad , x > \mu \quad F_x(x) = 1 - e^{-(x-\mu)} \quad \rightarrow P(X > x) = e^{-(x-\mu)}$$

$$f_{x(n)}(x) = n(e^{-(x-\mu)})^{n-1} e^{-(x-\mu)} = n e^{-n(x-\mu)} \quad x > \mu$$

$$(a.) \quad P(|X_{(1)} - \mu| \geq \epsilon) = 1 - P(|X_{(1)} - \mu| < \epsilon) = 1 - P(-\epsilon < X_{(1)} - \mu < \epsilon)$$

$$= 1 - P(\mu - \epsilon < X_{(1)} < \mu + \epsilon) = 1 - P(X_{(1)} < \mu + \epsilon)$$

$$= P(X_{(1)} > \mu + \epsilon) = [P(X > \mu + \epsilon)]^n = [e^{-(\mu + \epsilon - \mu)}]^n$$

$$= e^{-n\epsilon} \xrightarrow{n \rightarrow \infty} 0 \quad \rightarrow \quad X_{(1)} \xrightarrow{P} \mu$$

(b) Yes. $\frac{1}{Y_n} \xrightarrow{P} \frac{1}{\mu}$ since $Y_n \xrightarrow{P} \mu$ and the reciprocal function is continuous for $x > \mu$.